
Integrable Systems — Exercise Sheet 0

Why integrability?

Variant: Questions only

This sheet assumes only what Lecture 0 assumes: calculus, linear algebra, ordinary differential equations, and mechanics in Hamiltonian form, $\dot{q} = \partial H / \partial p$, $\dot{p} = -\partial H / \partial q$. A quantity $F(q, p)$ is conserved when $\dot{F} = 0$ along every solution.

Exercise 1 (the Hooke chain). Take N unit masses on a line, joined by unit springs, with both ends fixed: with x_n the displacement of mass n and $x_0 = x_{N+1} = 0$,

$$H = \frac{1}{2} \sum_{n=1}^N p_n^2 + \frac{1}{2} \sum_{n=0}^N (x_{n+1} - x_n)^2.$$

(i) Write Hamilton's equations, and show that $x_n(t) = \sin(nk\pi/(N+1)) \cos \omega t$, for $k \in \{1, \dots, N\}$, solves them exactly when $\omega = \omega_k = 2 \sin(k\pi/(2(N+1)))$, the frequencies of the lecture. (ii) Show that the vectors $s_k = (\sin(nk\pi/(N+1)))_{n=1}^N$ are orthogonal, with $|s_k|^2 = (N+1)/2$. With $Q_k = \sqrt{2/(N+1)} \sum_n \sin(nk\pi/(N+1)) x_n$ and P_k the same combination of the p_n , show that $H = \sum_k \frac{1}{2} (P_k^2 + \omega_k^2 Q_k^2)$ and that each mode energy $E_k = \frac{1}{2} (P_k^2 + \omega_k^2 Q_k^2)$ is conserved: N conserved quantities. (iii) For $N = 2$, write E_1 and E_2 in terms of x_1, x_2, p_1, p_2 and check directly from Hamilton's equations that each is conserved. (iv) For $N = 32$, the chain of the lecture's figure, compute ω_1 , the period of mode 1, and how many of its periods fit into $t = 10^4$. Compute $\omega_2/\omega_1, \omega_3/\omega_1$ and ω_4/ω_1 .

Exercise 2 (collisions with unequal masses). Two particles of masses $m_1, m_2 > 0$ collide elastically on a line, particle 1 on the left, with velocities $v_1 > v_2$ before and w_1, w_2 after. (i) Solve the conservation of momentum and of energy for (w_1, w_2) . Show that the solutions are $(w_1, w_2) = (v_1, v_2)$ and

$$w_1 = \frac{(m_1 - m_2)v_1 + 2m_2v_2}{m_1 + m_2}, \quad w_2 = \frac{(m_2 - m_1)v_2 + 2m_1v_1}{m_1 + m_2}.$$

From now on suppose the particles cannot pass through each other, so that particle 1 stays on the left. Show that this rules out the first solution and leaves the second. (ii) For the second solution, show that the centre-of-mass velocity is unchanged and the relative velocity reverses: $w_2 - w_1 = -(v_2 - v_1)$. (iii) For the second solution, show that $\{w_1, w_2\} = \{v_1, v_2\}$ as sets only if $m_1 = m_2$. (iv) A particle of mass 1 moving at velocity 1 hits a particle of mass 3 at rest. Find the velocities after the collision, and compare with two equal masses. What does this say about a one-dimensional gas of two species?

Exercise 3 (counting unknowns against laws). (i) Two particles of masses m_1, m_2 collide elastically in d dimensions. Count the unknown outgoing velocity components and the conservation laws. With $M = m_1 + m_2$, the centre-of-mass velocity V and the relative velocity $u = v_1 - v_2$, show that the kinetic energy is $\frac{1}{2}M|V|^2 + \frac{1}{2}\mu|u|^2$ with $\mu = m_1m_2/M$, and deduce that the possible outcomes are exactly the relative velocities u' with $|u'| = |u|$. What is this set for $d = 1$, $d = 2$ and $d = 3$ when the particles have a relative velocity, $u \neq 0$, and what is it when $u = 0$? (ii) Now N unit masses on a line meet at once. Show that fixed total momentum P and energy E confine the outgoing momenta to the set with $\sum_i (p_i - P/N)^2 = 2E - P^2/N$ in

the plane $\sum_i p_i = P$, and that this contains every rearrangement of the incoming momenta. Show that it is a sphere of dimension $N - 2$ and radius $\sqrt{2E - P^2/N} > 0$ unless all incoming momenta are equal, when it is a single point. What is it for $N = 2$ and $N = 3$? (iii) The lecture's figure of three colliding particles has incoming momenta $(1.6, 0.3, -1.2)$, and prints the generic outcome as $(-0.94, -0.14, 1.78)$. Compute P , $2E = \sum p_i^2$ and the radius of the circle. Check that the printed outcome has the same P and, to the rounding of its last digit, the same $\sum p_i^2$, and that it is not a rearrangement of the incoming momenta.

Exercise 4 (the FPUT spring and the Toda spring). The lecture's tutorial matches the Toda spring $V_T(r) = \frac{a}{b} e^{-br} + ar - \frac{a}{b}$ to the FPUT spring $\kappa(\frac{1}{2}r^2 + \frac{\alpha}{3}r^3)$ through cubic order, with $b = -2\alpha$ and $a = -\kappa/(2\alpha)$, for $\alpha \neq 0$ and $\kappa > 0$. (i) Expand V_T through order r^4 . (ii) Show that with the matched constants $V_T(r) - \kappa(\frac{1}{2}r^2 + \frac{\alpha}{3}r^3) = \frac{1}{6}\kappa\alpha^2r^4 + O(r^5)$. (iii) Show that the matched Toda spring satisfies $V_T(r) \geq 0$ for every r , with equality only at $r = 0$. (iv) Take $\kappa = 1$. Show that the FPUT spring has, besides its minimum at $r = 0$, a local maximum at $r = -1/\alpha$, of height $1/(6\alpha^2)$, and that it decreases without bound beyond it. For the lecture's $\alpha = \frac{1}{4}$, where is the barrier and how high is it? What happens to a single FPUT spring given more energy than that?

Exercise 5 (counting conserved quantities in quantum mechanics). Let H be a Hermitian $n \times n$ matrix with distinct eigenvalues $\lambda_1, \dots, \lambda_n$, and P_k the orthogonal projector onto the eigenline of λ_k . (i) Show that $HP_k = P_kH$, that $P_jP_k = \delta_{jk}P_k$, and that $P_1, \dots, P_n$ are linearly independent: every such H has n commuting, linearly independent conserved quantities. (ii) Show that $P_1, \dots, P_n$ span the same space of matrices as $1, H, \dots, H^{n-1}$. (iii) For $H = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}$ with a, b real and $b \neq 0$, find $P_{\pm}$ explicitly. (iv) What do (i) and (ii) say about defining a quantum integrable system as one with as many commuting conserved quantities as the dimension, or as degrees of freedom?

Exercise 6 ★. *The Toda chain keeps the eigenvalues of a matrix.* The open Toda chain of N unit masses has

$$H = \frac{1}{2} \sum_{n=1}^N p_n^2 + \sum_{n=1}^{N-1} e^{q_n - q_{n+1}},$$

so $\dot{q}_n = p_n$ and $\dot{p}_n = e^{q_{n-1} - q_n} - e^{q_n - q_{n+1}}$, the terms with q_0 and q_{N+1} absent. Its Lax matrix, which Lecture 8 builds, is the symmetric tridiagonal matrix L with $p_1, \dots, p_N$ on the diagonal and $a_n = e^{(q_n - q_{n+1})/2}$ beside it. (i) For $N = 2$, show directly from Hamilton's equations that $\text{tr } L$ and $\det L$ are conserved, and conclude that the eigenvalues of L are. (ii) Two Toda particles that collide separate again: $q_2 - q_1 \rightarrow +\infty$ as $t \rightarrow \pm\infty$, and their momenta tend to limits (take this as given: Lecture 12 proves it for two particles by solving the motion, and cites it for N). Deduce that the outgoing momenta are the incoming ones, exchanged. (iii) For $N = 3$, show that $\text{tr } L^3 = \sum_n p_n^3 + 3 \sum_{n=1}^2 e^{q_n - q_{n+1}}(p_n + p_{n+1})$ and that it is conserved. Why does it follow, with $\text{tr } L = \sum_n p_n$ and $\text{tr } L^2 = 2H$, that the eigenvalues of L are conserved?